````
---
title: "MRI section"
output: html_document
date: "2026-09-09"
---

```{r setup, include=FALSE}
knitr::opts_chunk$set(echo = TRUE)
```

#Import libraries
```{r}

library(psych)
library(readr)
library(dplyr)
library(ggplot2)
library(tidyr)
library(tidyselect)
library(tidyverse)
library(janitor)
library(stringr)
library(readxl)
library(forcats)
library(brms)
library(knitr)
library(bayestestR)
library(emmeans)

```




# GPi vs STN FA differences at baseline?
Stats here

```{r}

#import DTI data
newMRI_cleaned <- read_excel("DTImeasures_processed_long26May2026.csv")

 # FA data only

 newMRI_cleanedFA <- newMRI_cleaned %>%
   filter(type=="FA")

 # MD data only

 newMRI_cleanedMD <- newMRI_cleaned %>%
   filter(type=="MD")

#add in demog info - confidential demog info of PD participants
quick_fix <- PD_onerow %>%
  dplyr::select(subjID, DBS_location) %>%
  mutate(subjID = as.character(subjID))

newMRI_longPE <- newMRI_cleaned %>%
  left_join(quick_fix, by = "subjID")

newMRI_longPE$DBS_location <- fct_relevel(newMRI_longPE$DBS_location, "STN", "GPi")


library(brms)

FA_stats <- newMRI_longPE %>%
  filter(DBS_location %in% c("STN", "GPi")) %>%
  filter(type=="FA")
````
````

```
# Run ONE model with Region as a predictor and an interaction with Group
# This naturally accounts for the multiple regions without needing post-hoc correction
fMRImodel_FA <- brms::brm(
  formula = Value ~ DBS_location * Tract + Side + (1 + Tract | subjID),
  data = FA_stats,
  family = student(), # or whichever distribution fits your data
  chains = 4, cores = 4
)

summary(fMRImodel_FA)


library(emmeans)

# 1. Calculate means for each group WITHIN each region
# This tells emmeans: "Group by Regions, then compare DBS_location"
# Because 'Side' isn't mentioned, emmeans averages it out completely.
region_comparisons <- emmeans(fMRImodel_FA, ~ DBS_location | Tract)

# 2. Run the pairwise test (STN vs GPi) for each region
# This is the "Hypothesis Test" you need
group_diffs_per_region <- pairs(region_comparisons)

# 3. View the results
summary(group_diffs_per_region)

library(bayestestR)


# 1. Get the standard summary (Estimates and CIs)
diff_summary <- summary(group_diffs_per_region)

# 2. Get the pd results and convert to a data frame
pd_vals <- as.data.frame(p_direction(group_diffs_per_region))

# 3. Combine them into one clean table
final_table <- cbind(diff_summary, pd = pd_vals$pd)

# 4. View the result
print(final_table)




```


#GPi vs STN MD differences at baseline?
Stats here
```{r}


library(brms)

MD_stats <- newMRI_longPE %>%
  filter(DBS_location %in% c("STN", "GPi")) %>%
  filter(type=="MD")

#need to re-scale MD values - model won't converge otherwise

# Multiply MD values by 1000 to make them easier for the model to "see"
MD_stats$Value_scaled <- MD_stats$Value * 1000


# Run ONE model with Region as a predictor and an interaction with Group
# This naturally accounts for the multiple regions without needing post-hoc correction
fMRImodel_MD <- brms::brm(
  formula = Value_scaled ~ DBS_location * Tract + Side + (1 + Tract | subjID),
```

```
    data = MD_stats,
    family = student(), # or whichever distribution fits your data
    chains = 4, cores = 4
  )

  summary(fMRImodel_MD)


  library(emmeans)

  # 1. Calculate means for each group WITHIN each region
  # This tells emmeans: "Group by Regions, then compare DBS_location"
  # Because 'Side' isn't mentioned, emmeans averages it out completely.
  region_comparisons <- emmeans(fMRImodel_MD, ~ DBS_location | Tract)

  # 2. Run the pairwise test (STN vs GPi) for each region
  # This is the "Hypothesis Test" you need
  group_diffs_per_region <- pairs(region_comparisons)

  # 3. View the results
  summary(group_diffs_per_region)

  library(bayestestR)







  # 1. Get the standard summary (Estimates and CIs)
  diff_summary <- summary(group_diffs_per_region)

  # 2. Get the pd results and convert to a data frame
  pd_vals <- as.data.frame(p_direction(group_diffs_per_region))

  # 3. Combine them into one clean table
  final_table <- cbind(diff_summary, pd = pd_vals$pd)

  # 4. View the result
  print(final_table)



  ```


  # FA & DDM metrics
  Stats here

  ```{r}

  FA_drift <- read_csv("FA_drift_29May2026.csv")


  modelFAPreOFF <- brms::brm(Value ~ Tract*DBS_location*v_Shift_OFF_minus_Pre + Side + (1 + Tract
  | subjID), family= student(), data = FA_drift, chains = 4, cores = 4)
  summary(modelFAPreOFF)

  # Load necessary libraries
  library(emmeans)
  library(bayestestR)

  # 1. Extract the marginal trends (slopes) for the shift variable.
  # The "|" symbol tells the function to look at Regions nested within each DBS group.
  shift_slopes <- emtrends(modelFAPreOFF,
                           specs = ~ Tract | DBS_location,
                           var = "v_Shift_OFF_minus_Pre")

  slopes_summary <- as.data.frame(summary(shift_slopes, point.est = median, level = 0.95, hpd =
```

```
TRUE))
pd_results <- as.data.frame(p_direction(shift_slopes))
final_summary_table <- slopes_summary %>%
  mutate(pd = pd_results$pd) %>%
  mutate(pd_formatted = sprintf("%.2f%%", pd * 100)) %>%
  dplyr::select(DBS_location, Tract, v_Shift_OFF_minus_Pre.trend, lower.HPD, upper.HPD,
pd_formatted)
final_summary_table
kable(final_summary_table, format = "markdown", row.names = FALSE)


###############


modelFAOFF_ON2 <- brms::brm(Value ~ Tract*DBS_location*v_Shift_ON_minus_OFF + Side + (1 + Tract
| subjID), family= student(), data=FA_drift, chains = 4, cores = 4)
summary(modelFAOFF_ON2)



# Load necessary libraries
library(emmeans)
library(bayestestR)

# 1. Extract the marginal trends (slopes) for the shift variable.
# The "|" symbol tells the function to look at Regions nested within each DBS group.
shift_slopes <- emtrends(modelFAOFF_ON2,
                         specs = ~ Tract | DBS_location,
                         var = "v_Shift_ON_minus_OFF")

slopes_summary <- as.data.frame(summary(shift_slopes, point.est = median, level = 0.95, hpd =
TRUE))
pd_results <- as.data.frame(p_direction(shift_slopes))
final_summary_table <- slopes_summary %>%
  mutate(pd = pd_results$pd) %>%
  mutate(pd_formatted = sprintf("%.2f%%", pd * 100)) %>%
  dplyr::select(DBS_location, Tract, v_Shift_ON_minus_OFF.trend, lower.HPD, upper.HPD,
pd_formatted)
final_summary_table
kable(final_summary_table, format = "markdown", row.names = FALSE)



```

#MD & DDM metrics
```{r}

MD_drift <- read_csv("MD_drift_29May2026.csv")


modelMDPreOFF <- brms::brm(Value_scaled ~ Tract*DBS_location*v_Shift_OFF_minus_Pre + Side + (1
+ Tract | subjID), family= student(), data = MD_drift, chains = 4, cores = 4)
summary(modelMDPreOFF)

# Load necessary libraries
library(emmeans)
library(bayestestR)

# 1. Extract the marginal trends (slopes) for the shift variable.
# The "|" symbol tells the function to look at Regions nested within each DBS group.
shift_slopes <- emtrends(modelMDPreOFF,
                         specs = ~ Tract | DBS_location,
                         var = "v_Shift_OFF_minus_Pre")

slopes_summary <- as.data.frame(summary(shift_slopes, point.est = median, level = 0.95, hpd =
TRUE))
pd_results <- as.data.frame(p_direction(shift_slopes))
final_summary_table <- slopes_summary %>%
```

```
    mutate(pd = pd_results$pd) %>%
    mutate(pd_formatted = sprintf("%.2f%%", pd * 100)) %>%
    dplyr::select(DBS_location, Tract, v_Shift_OFF_minus_Pre.trend, lower.HPD, upper.HPD,
  pd_formatted)
  final_summary_table
  kable(final_summary_table, format = "markdown", row.names = FALSE)



  ###############


  modelMDOFF_ON <- brms::brm(Value_scaled ~ Tract*DBS_location*v_Shift_ON_minus_OFF + Side + (1 +
  Tract | subjID), family= student(), data=MD_drift, chains = 4, cores = 4)
  summary(modelMDOFF_ON)

  # Load necessary libraries
  library(emmeans)
  library(bayestestR)

  # 1. Extract the marginal trends (slopes) for the shift variable.
  # The "|" symbol tells the function to look at Regions nested within each DBS group.
  shift_slopes <- emtrends(modelMDOFF_ON,
                           specs = ~ Tract | DBS_location,
                           var = "v_Shift_ON_minus_OFF")

  slopes_summary <- as.data.frame(summary(shift_slopes, point.est = median, level = 0.95, hpd =
  TRUE))
  pd_results <- as.data.frame(p_direction(shift_slopes))
  final_summary_table <- slopes_summary %>%
    mutate(pd = pd_results$pd) %>%
    mutate(pd_formatted = sprintf("%.2f%%", pd * 100)) %>%
    dplyr::select(DBS_location, Tract, v_Shift_ON_minus_OFF.trend, lower.HPD, upper.HPD,
  pd_formatted)
  final_summary_table
  kable(final_summary_table, format = "markdown", row.names = FALSE)







  ```


  #VTA centroids & DDM metrics
  Import data
  ```{r}

  teed_VTAcoords <- read_excel("teed_VTAcoords.xlsx")


  HDDM_SubjectLevel_Estimates <-
  read_csv("HDDM_SubjectLevel_Estimates_Unified_mD2TbF_29May2026.csv")

  HDDM_SubjectLevel_Estimates$subjID <- as.character(HDDM_SubjectLevel_Estimates$subjID)
  teed_VTAcoords$subjID <- as.character(teed_VTAcoords$subjID)


  #preprocess data

  teed_VTAcoords <- teed_VTAcoords %>%
    mutate(subjID = str_remove(subjID, "sub-"))


  #join

  VTA_behav <- teed_VTAcoords %>%
    left_join(HDDM_SubjectLevel_Estimates, by = "subjID")
```

```
#various reasons - e.g. post op CT done at another hospital/no access to CT etc.

VTA_behav_clean <- VTA_behav %>%
  filter(!is.na(DBS_location)) %>%
  filter(!is.na(L_TEED))


#clean up variable names etc to create a Side column (put data into long format)


# Reshape the data to long format
VTA_behav_clean_long <- VTA_behav_clean %>%
  pivot_longer(
    # FIX: Only pivot columns that explicitly start with "L_" or "R_"
    cols = matches("^([LR])_"),

    names_to = c("Side", ".value"),
    names_pattern = "^([LR])_(.*)"
  ) %>%
  # Change "L" and "R" to "Left" and "Right"
  mutate(Side = case_when(
    Side == "L" ~ "Left",
    Side == "R" ~ "Right",
    TRUE ~ Side
  ))


# View the result
head(VTA_behav_clean_long)


#split into groups for modeling


VTA_behav_clean_longSTN <- VTA_behav_clean_long %>%
  filter(DBS_location=="STN")

VTA_behav_clean_longGPi <- VTA_behav_clean_long %>%
  filter(DBS_location=="GPi")



```


STN group

```{r}

modelSTNm <- brm(
  formula = mvbind(X_distance_from_midline, VTA_MNI_Y, VTA_MNI_Z) ~ v_Shift_OFF_minus_Pre +
Side + (1 | subjID),
  data = VTA_behav_clean_longSTN,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000
)
summary(modelSTNm)

pd(modelSTNm)

#
modelSTNm2 <- brm(
  formula = mvbind(X_distance_from_midline, VTA_MNI_Y, VTA_MNI_Z) ~ v_Shift_ON_minus_OFF + Side
+ (1 | subjID),
  data = VTA_behav_clean_longSTN,
  family = student(),
```

```
  chains = 4,
  iter = 2000,
  warmup = 1000
)
summary(modelSTNm2)

pd(modelSTNm2)
```

##GPi group

```{r}

#
modelGPim <- brm(
  formula = mvbind(X_distance_from_midline, VTA_MNI_Y, VTA_MNI_Z) ~ v_Shift_OFF_minus_Pre +
Side + (1 | subjID),
  data = VTA_behav_clean_longGPi,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000
)
summary(modelGPim)

pd(modelGPim)

#
modelGPim2 <- brm(
  formula = mvbind(X_distance_from_midline, VTA_MNI_Y, VTA_MNI_Z) ~ v_Shift_ON_minus_OFF + Side
+ (1 | subjID),
  data = VTA_behav_clean_longGPi,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000
)
summary(modelGPim2)

pd(modelGPim2)
```

#Apathy & DDM metrics (Not MRI here)
Are DDM shifts associated with apathy status and trajectory?

```{r}

apathy_drift <- read_csv("apathy_drift_29May2026.csv")

######################### adjusted models - with DBS_location in the models


model_robust <- brm(
  formula = v_Shift_OFF_minus_Pre ~ apatheticF + DBS_location,
  data = apathy_drift,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  cores = 4
)

summary(model_robust)
pd(model_robust)
```

```
model_robust2 <- brm(
  formula = v_Shift_ON_minus_OFF ~ apatheticF + DBS_location ,
  data = apathy_drift,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  cores = 4
)

summary(model_robust2)
pd(model_robust2)


###############


# Get the pd for your first model
p_direction(model_robust)

# Get the pd for your second model
p_direction(model_robust2)


################## un adjusted models

model_robustU <- brm(
  formula = v_Shift_OFF_minus_Pre ~ apatheticF,
  data = apathy_drift,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  cores = 4
)

summary(model_robustU)
pd(model_robustU)

model_robust2U <- brm(
  formula = v_Shift_ON_minus_OFF ~ apatheticF,
  data = apathy_drift,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  cores = 4
)

summary(model_robust2U)
pd(model_robust2U)

########################## apathy status


model_robustAS <- brm(
  formula = v_Shift_OFF_minus_Pre ~ apathy_status + DBS_location,
  data = apathy_drift,
  family = student(),
  chains = 4,
  iter = 2000,
  warmup = 1000,
  cores = 4
)

summary(model_robustAS)
pd(model_robustAS)

model_robust2AS <- brm(
```

```
    formula = v_Shift_ON_minus_OFF ~ apathy_status + DBS_location ,
    data = apathy_drift,
    family = student(),
    chains = 4,
    iter = 2000,
    warmup = 1000,
    cores = 4
  )

  summary(model_robust2AS)
  pd(model_robust2AS)



  ```
```
```